Ana Aquiles

Data Scientist · Computational Neuroscience · Deep Learning · Machine Learning· Python

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PROFESSIONAL SUMMARY

I am a data scientist with a PhD specialized in Computational Neuroscience and 5**+ years of experience** converting large, high-dimensional time-series and imaging datasets into actionable insights. I specialize in building end-to-end Python pipelines using machine learning, deep learning, and statistical methods. My experience spans various data domains, including neural signals, movie recordings, MRI volumes, and stochastic time series, allowing me to apply a *domain-agnostic approach with techniques such as dimensional reduction and anomaly and pattern detection*. I am skilled at managing projects from data design through to modeling and effectively communicating findings to non-technical stakeholders.

CORE TECHNICAL SKILLS

Machine & deep learning: Supervised/unsupervised learning, deep learning (**PyTorch, TensorFlow**), dimensionality reduction, classification & clustering, multivariate time-series forecasting, anomaly and pattern detection

Statistics & theory: Statistical modelling, population/state-space geometry, information theory, geometric & topological data analysis, generative null models for significance testing

LLMs: Claude API integrated into research workflows for code development acceleration and synthetic-data generation

Data & engineering: **Python** (advanced), **NumPy**, **SciPy**, **pandas, scikit-learn**, **OpenCV**, **NiBabel, SQL, ETL**/data pipelines, **Git/GitHub, Unix/Linux**, reproducible and open science

Visualisation & communication: communicating results to non-technical stakeholders; mentoring and workshops

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher - Computational Neuroscience **Apr 2024 - Mar 2026**

CNRS, Institut du Génomique Fonctionnelle, Montpellier, France.

- Built ML and deep-learning pipelines in Python to process large-scale, high-frequency time-series and video datasets (>10 GB per experiment), engineering the full path from raw signal to model-ready features.
- Applied dimensionality-reduction and population/state-space geometry methods to uncover directed coupling and bistable dynamics in multivariate time-series — a technique directly transferable to anomaly detection and regime-change forecasting in any complex sequential dataset (published, bioRxiv 2026).
- Integrated Claude (Anthropic LLM API) into the research workflow to accelerate deep-learning code development and to design a null-generative model producing synthetic data for statistical significance testing, strengthening the rigor of model validation.
- Mentored team members on Python and ML best practices; led internal workshops on applied machine-learning methods.

Research Data Engineer - Computational Neuroscience
CNRS, Institut du Génomique Fonctionnelle, Montpellier, France. **Sep 2023 - Apr 2024**

- Designed and implemented data pipelines for high-throughput, multi-channel imaging data, including quality-control and batch-processing scripts to guarantee integrity at scale.
- Partnered across institutions to deliver standardised, reproducible data workflows adopted by multiple research teams.

PhD Researcher - Computational & Systems Data Analyst **Aug 2019 - Dec 2023**
UNAM, Neurobiology Institute. Brain Connectivity Laboratory, Queretaro, Mexico.

- Designed and built a full analytical pipeline integrating three distinct time-series/imaging modalities (electrophysiology, local field potentials, MRI) into a single comparable feature space — published in Scientific Reports (2023) and iScience (2025).
- Applied population-dynamics, machine-learning, and information-theory methods to high-dimensional electrophysiological time-series as part of a cross-institutional collaboration (visiting researcher, 2023).
- Supervised and trained 4 MSc/BSc students on data-analysis projects; co-authored 5 peer-reviewed publications.

Research Assistant - Computational & Systems Data Analyst **Jan 2018 - Aug 2019**
UNAM, Neurobiology Institute. Brain Connectivity Laboratory, Queretaro, Mexico.

- Led data acquisition and analysis for imaging/diffusion-tensor studies; built automated processing scripts in Python.
- Trained postgraduate students in image-analysis workflows and experimental protocol design.

EDUCATION

PhD in Biomedical Sciences
Neurobiology Institute, UNAM CAMPUS, Juriquilla. Queretaro | 08/2019 - 12/2023
Bachelor of Science in Biology
Faculty of Sciences, UNAM, Mexico | 08/2014 - 01/2018

KEY TRAINING AND CERTIFICATIONS

L'apprentissage automatique (machine learning) appliqué à l'analyse- BIOCAMPUS **Jun 2025**
Postdoctoral Scientist

ib Cegos. Python Perfectionemment- CNRS Formation **May 2025**
Postdoctoral Scientist

NEUBIAS workshop in bioimage analysis at Pasteur Institute, France **May 2025**
Postdoctoral Scientist
Computational methods and application into segmentation and image treatment using Python

Neuromatch Computational Neuroscience school, online **Jul 2020**
Ph.D student.

LASCON (Latin American School on Computational Neuroscience), Brazil **Jan 2020**

UV, University of Valparaiso. Fundacion Ciencia y Vida. Chile **Dec 2016- Feb 2017**
BS. in Biology, Internship
Data Analysis and neurocomputation at Computational Neuroscience Lab lead by Dr. Patricio Orio

SELECTED PUBLICATIONS

- Aquiles A. et al. Assessing the degree of cortical dyslamination through electrical pattern analysis. iScience (2026).
- Aquiles A. et al. Altered functional connectivity and network excitability in a model of cortical dysplasia. Scientific Reports 13 (2023).
- Santiago-Andres Y, Aquiles A, et al. Intracellular calcium signaling and tumor recurrence in non-functioning pituitary adenomas. Int J Mol Sci 25 (2024).
- Aquiles A. et al. Population geometry reveals directed coupling and transient bistability in spontaneous pituitary secretion. bioRxiv (2026), under review in *Nature Communications Physics*.
- **Full list: [Google Scholar / ORCID link]**

LANGUAGES

Spanish - native **English**- proficient **French**- proficient **Portuguese**- intermediate